

EXP-13- EMBEDDED C PROGRAM TO ADD AN ARRAY OF 16-BIT NUMBERS AND STORE
THE 32-BIT RESULT IN INTERNAL RAM

CODE:

```
#include <reg51.h>
```

```
void main(void)
```

```
{
```

```
    unsigned int i;
```

```
    unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
```

```
    unsigned long sum = 0;
```

```
    for (i = 0; i < 5; i++)
```

```
    {
```

```
        sum = sum + array[i];
```

```
    }
```

```
    P0 = (unsigned char)(sum & 0xFF);    // Lower 8 bits
```

```
    P1 = (unsigned char)((sum >> 8) & 0xFF); // Next 8 bits
```

```
    P2 = (unsigned char)((sum >> 16) & 0xFF); // Upper 8 bits
```

```
    while(1);
```

```
}
```

CIRCUIT DIAGRAM:

The screenshot displays a Verilog HDL simulation environment. The main window shows the Verilog code for a module named `testbench`. The code defines a module `testbench` with inputs `clk` and `rst`, and outputs `PO`, `P1`, and `P2`. The code includes a `reg` declaration for `sum` and a `for` loop that iterates over an array of values. The code is as follows:

```

module testbench (
    input clk,
    input rst,
    output PO,
    output P1,
    output P2
);
    reg sum;

    for (i = 0; i < 10; i++)
    begin
        sum = sum + array[i];
    end

    PO = sum;
    P1 = sum;
    P2 = sum;
endmodule

```

The simulation results are shown in the bottom panel, which includes a waveform viewer and a console window. The waveform viewer displays the signals `clk`, `rst`, `PO`, `P1`, and `P2`. The console window shows the simulation output, including the values of the signals and the execution time.

The circuit diagram is shown in the top right corner, illustrating the internal structure of the module. It includes a clock input `clk`, a reset input `rst`, and three output ports labeled `PO`, `P1`, and `P2`. The diagram shows the internal logic blocks and their connections, including a clock divider and a reset logic block.